

# Directional plasmon emission in a mixed dielectric-plasmonic structure: Application to chiral sensing.

Guillermo Serrera<sup>1</sup>, Javier González-Colsa<sup>1</sup>, Vincenzo Giannini<sup>2</sup>, José M. Saiz<sup>1</sup> and Pablo Albella<sup>1</sup>\*

<sup>1</sup> Department of Applied Physics, University of Cantabria, Avda. Los Castros, s/n, 39005 Santander, Spain

<sup>2</sup> Instituto de Estructura de la Materia (IEM), Consejo Superior de Investigaciones Científicas (CSIC), Serrano 121, 28006 Madrid, Spain.

[\\*pablo.albella@unican.es](mailto:pablo.albella@unican.es)

The concept of chirality, related to objects and structures which present non-superimposable mirror images in any plane of symmetry, has attracted a lot of attention recently in research fields like life sciences, where for example, it has been suggested as the underlying cause of Parkinson's and Alzheimer's diseases [1]. Enantiomers or optical isomers are chiral molecules which can have different effects as drugs, and therefore their distinction is relevant in the pharmacological industry, where single-enantiomer drugs are being developed to improve their effectiveness and compatibility [2].

In nanophotonics, intensive research is being performed to develop novel structures and devices capable of sensing enantiomeric proportions by enhancing the intrinsically weak Circular Dichroism (CD) optical effect. Our work focuses on the use of chiral structures capable of enhancing this effect. Special attention is paid to the distribution of near-field optical chirality, seeking for homogeneous high-dissymmetry factor areas available to analytes [3].

Here I will present the results of a recent study where, using computational FDTD methods, we demonstrate the capabilities of a High Refractive Index Dielectric (HRID) metasurface to offer both a high far field enhancement of the CD effect and a strong near-field chiral enhancement. This HRID metasurface is then set on a gold layer to support directional Surface Plasmon Polariton (SPP) generation [4] [5], thus obtaining a directional plasmonic chiral sensor device [6].

## REFERENCES

- [1] C.M. Dobson, Protein folding and misfolding, *Nature (London)* **426**, 884, 2003.
- [2] J. McConathy and J. Owens, Stereochemistry in drug action, *Prim. Care Companion J. Clin. Psychiatry* **5**, 70, 2003.
- [3] F. Reyes Gómez, O. N. Oliveira, Jr., P. Albella and J. R. Mejía-Salazar, "Enhanced chiroptical activity with slotted high refractive index dielectric nanodisks", *Physical Review B* **101**, 155403, 2020.
- [4] J. González-Colsa, G. Serrera, J. M. Saiz, F. González, F. Moreno and P. Albella, "On the performance of a tunable grating-based high sensitivity unidirectional plasmonic sensor", *Optics Express* **29**, 9, 13733-13745, 2021.
- [5] J. Lin et al., "Polarization-controlled tunable directional coupling of surface plasmon polaritons," *Science* **340**, 6130, 331–334, 2013.
- [6] G. Serrera, J. González-Colsa, V. Giannini, J. M. Saiz and P. Albella, "Enhanced chiral sensing via directional emission of Surface Plasmon Polaritons" (under review).